

WILP, BITS Pilani

Session 15: Gaussian Mixture Models & Practice Problems

Course: Introduction to Statistical Methods (ISM) | Instructor: Dr. YVK Ravi Kumar

1 Gaussian Mixture Models (GMM)

Model complex multimodal data by combining several weighted Gaussian distributions.

A Gaussian Mixture Model represents an overall population probability density as a linear combination of K underlying Gaussian components:

$$p(x) = \sum_{k=1}^K \pi_k \mathcal{N}(x | \mu_k, \Sigma_k) \quad (1)$$

where π_k are mixing coefficients satisfying $\sum_{k=1}^K \pi_k = 1$ and $\pi_k \geq 0$.

Everyday Picture & Intuition

Suppose share prices of Company A average \$90 and Company B average \$100. When combined into one pool, the price distribution shows two distinct peaks. GMM models this combined density as a weighted blend of two bell curves.

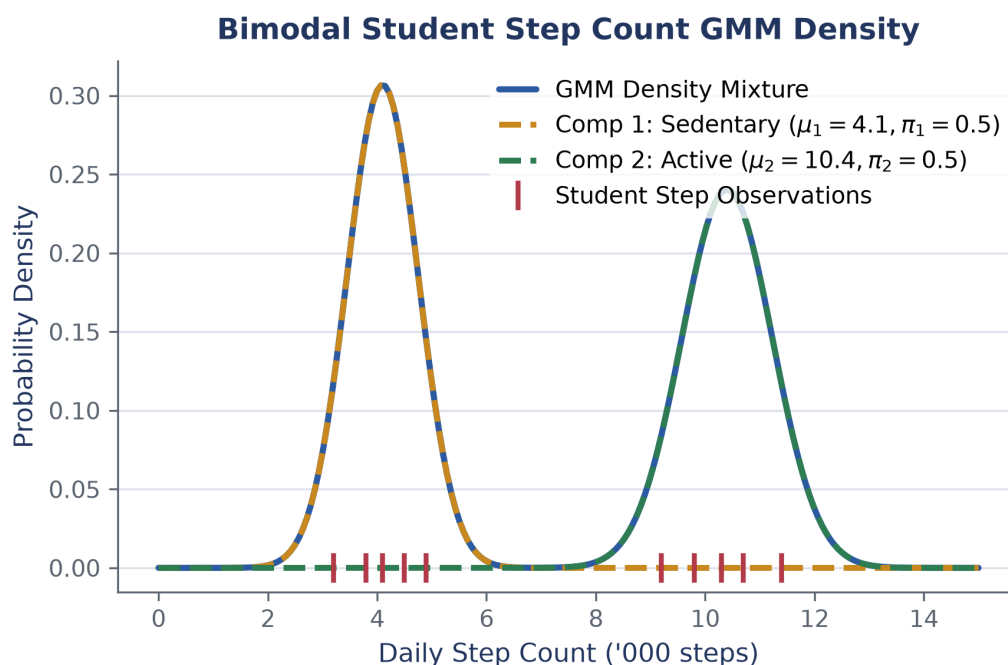


Figure 1: Bimodal student step count density modeled using a 2-component Gaussian Mixture Model.

2 The Expectation-Maximization (EM) Algorithm

Iteratively estimate component parameters and soft cluster responsibilities.

Because cluster assignments are unobserved latent variables $z_k \in \{0, 1\}$, parameter likelihood cannot be solved in closed form. The Expectation-Maximization (EM) algorithm alternates between two steps until convergence:

2.1 E-Step (Expectation)

Compute posterior responsibilities $\gamma(z_k)$ using Bayes' theorem:

$$\gamma(z_k) = P(z_k = 1 \mid x) = \frac{\pi_k \mathcal{N}(x \mid \mu_k, \Sigma_k)}{\sum_{j=1}^K \pi_j \mathcal{N}(x \mid \mu_j, \Sigma_j)} \quad (2)$$

2.2 M-Step (Maximization)

Re-estimate component parameters using current responsibilities $N_k = \sum_{i=1}^N \gamma(z_{ik})$:

$$\mu_k^{\text{new}} = \frac{1}{N_k} \sum_{i=1}^N \gamma(z_{ik}) x_i \quad (3)$$

$$\Sigma_k^{\text{new}} = \frac{1}{N_k} \sum_{i=1}^N \gamma(z_{ik}) (x_i - \mu_k^{\text{new}})(x_i - \mu_k^{\text{new}})^T \quad (4)$$

$$\pi_k^{\text{new}} = \frac{N_k}{N} \quad (5)$$

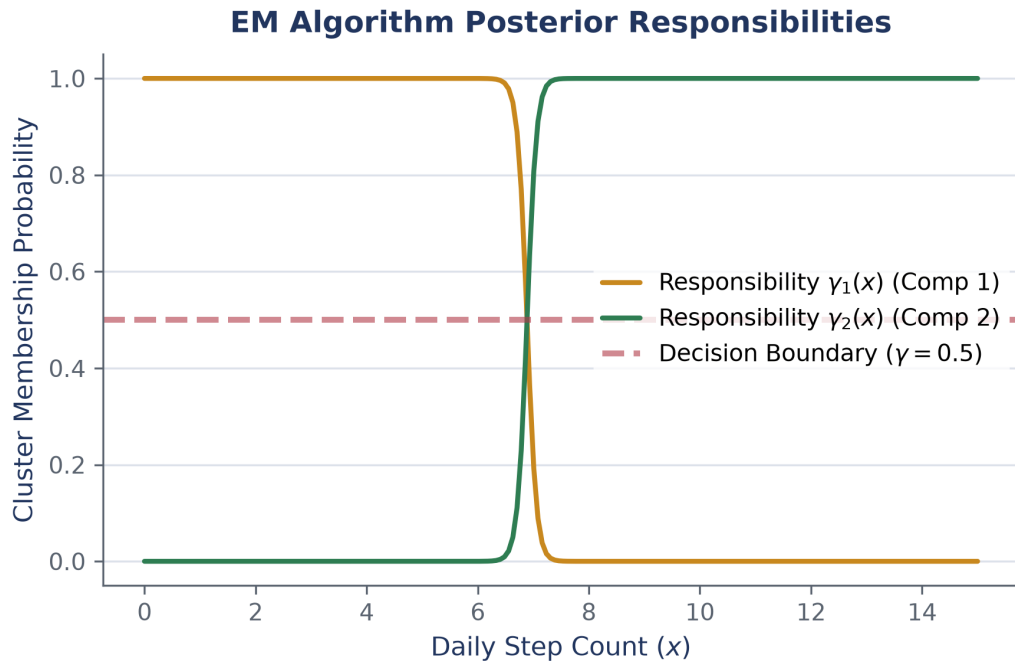


Figure 2: Soft cluster responsibilities $\gamma_k(x)$ assigning probabilities across data points.

Fully Worked Example

A wellness app logs step counts (in thousands) of 10 students: $\{3.2, 3.8, 4.1, 4.5, 4.9, 9.2, 9.8, 10.3, 10.7, 11.4\}$. Component 1 consists of the first 5 values (sedentary); Component 2 consists of the last 5 values (active).

Step 1: Calculate mixing coefficients:

$$\pi_1 = \frac{5}{10} = 0.50, \quad \pi_2 = \frac{5}{10} = 0.50 \quad (6)$$

Step 2: Calculate component means:

$$\begin{aligned} \mu_1 &= \frac{3.2 + 3.8 + 4.1 + 4.5 + 4.9}{5} = \frac{20.5}{5} = 4.10 \\ \mu_2 &= \frac{9.2 + 9.8 + 10.3 + 10.7 + 11.4}{5} = \frac{51.4}{5} = 10.28 \end{aligned}$$

Step 3: Calculate component variances $s^2 = \frac{\sum (x - \bar{x})^2}{n-1}$:

$$\begin{aligned} s_1^2 &= \frac{(-0.9)^2 + (-0.3)^2 + 0^2 + 0.4^2 + 0.8^2}{4} = \frac{1.70}{4} = 0.425 \\ s_2^2 &= \frac{(-1.08)^2 + (-0.48)^2 + 0.02^2 + 0.42^2 + 1.12^2}{4} = \frac{2.828}{4} = 0.707 \end{aligned}$$

Step 4: Formulate total GMM PDF:

$$p(x) = 0.50\mathcal{N}(x \mid 4.10, 0.425) + 0.50\mathcal{N}(x \mid 10.28, 0.707) \quad (7)$$

3 Proportion Hypothesis Testing Practice

Master core statistical tests: proportions and simple regression estimation.

Fully Worked Example

A company tested employee performance before and after training. Before: 400 out of 500 performed well ($p_1 = 0.800$). After: 440 out of 600 performed well ($p_2 = 0.7333$). Test for significant difference at $\alpha = 0.05$.

Step 1: Compute pooled proportion $\hat{p}$:

$$\hat{p} = \frac{400 + 440}{500 + 600} = \frac{840}{1100} = 0.7636, \quad \hat{q} = 1 - 0.7636 = 0.2364 \quad (8)$$

Step 2: Compute standard error:

$$SE = \sqrt{0.7636 \times 0.2364 \times \left(\frac{1}{500} + \frac{1}{600} \right)} = 0.0257 \quad (9)$$

Step 3: Calculate Z-statistic:

$$Z = \frac{0.8000 - 0.7333}{0.0257} = \frac{0.0667}{0.0257} = 2.59 \quad (10)$$

Since $|Z| = 2.59 > 1.96$, reject H_0 . There is a statistically significant difference in performance proportions.

Watch Out! Pitfalls

In GMM parameter estimation, initialization matters! Bad initial guesses for component means can cause the EM algorithm to get stuck in local optima.

Key Takeaways

- GMM models complex distributions as linear combinations of K Gaussians with mixing weights π_k .
- EM algorithm alternates between E-step (computing posterior responsibilities) and M-step (updating means and covariances).
- Soft clustering assigns probabilities rather than hard binary boundaries.